DHEERAJ KUMAR YADAV

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EDUCATION

Sharda University Greater Noida, India
Bachelor of Technology in Computer Science & Engineering (AI & Machine Learning) *Sep 2022 – Jul 2026*

- CGPA: 7.5/10.0

Kendriya Vidyalaya AMC, Lucknow Lucknow, India
Class XII – CBSE | Percentage: 75% (HSC) *2019 – 2020*

Kendriya Vidyalaya Fort William, Kolkata Kolkata, India
Class X – CBSE | Percentage: 70% (SSC) *2017 – 2018*

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL
ML/AI Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, VADER, TextBlob, YOLOv5
Web Technologies: HTML5, CSS3, JavaScript, REST API, Responsive Design
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, AWS
Data & Libraries: NumPy, Pandas, Matplotlib, MySQL, Data Preprocessing, Feature Engineering

EXPERIENCE

Web Development Intern Jun 2024 – Aug 2024
Techlightz Infosystem *Remote*

- Engineered a responsive, high-performance yoga platform using HTML5, CSS3, and JavaScript, driving a **40%** increase in mobile traffic and expanding the client's digital reach.
- Optimized website performance and Core Web Vitals, slashing page load speed by **35%** through code minification, aggressive image compression, and lazy loading.
- Ensured **100%** compliance with WCAG accessibility standards by executing rigorous cross-browser testing (Chrome, Firefox, Safari, Edge), reducing user bounce rate by **15%**.

PROJECTS

AI-Based Crop Disease Detection | *TensorFlow, Keras, ResNet50, OpenCV, Python* GitHub

- Developed an end-to-end CNN image classification pipeline using ResNet50 transfer learning to detect 10 crop disease categories, achieving **92% accuracy** and cutting agricultural yield risks by an estimated 20%.
- Accelerated model development by reducing training time by **60%** using GPU acceleration and specialized data augmentation on a dataset of 15,000+ images.
- Mitigated overfitting by **25%** through strategic regularization and dropout layers, ensuring high model generalization for real-world field deployments.

Insight-Stream: AI Spreadsheet Assistant | *GPT API, Pandas, REST API, JavaScript* GitHub

- Architected a web-based LLM application for automated data analysis, reducing manual data processing time by **70%** and saving users significant operational overhead.
- Engineered a pipeline processing datasets up to 100K rows with **95%** natural language query understanding accuracy, automating deep data cleaning and statistical visualization.
- Deployed automated trend, outlier, and correlation detection mechanisms via a scalable RESTful API, improving dashboard rendering efficiency by **30%**.

Music Recommendation System | *NLP, VADER, TextBlob, Python, REST API* GitHub

- Designed an emotion-driven NLP recommendation engine that boosted platform user engagement by **45%** through real-time mood and lyrical profiling.
- Achieved **88%** sentiment classification accuracy utilizing VADER and TextBlob libraries to dynamically index and categorize a database of 5,000+ songs.
- Integrated a real-time music streaming API into a responsive front-end interface, reducing streaming latency by **20%**.

Real-Time Object Detection | *YOLOv5, OpenCV, Python*

- Implemented a high-throughput YOLOv5 computer vision pipeline capable of detecting 80+ object classes in real-time at a continuous **30 FPS**.
- Optimized model architecture for edge-device deployment, cutting memory footprint by **50%** and reducing cloud compute hosting infrastructure costs.

CERTIFICATIONS & LEADERSHIP

- **Certifications:** AWS Academy Graduate – Foundation in Java (AWS) | Sustainable Development (NPTEL – IIT)
- **Leadership:** Event Manager – Ideathon: Directed a cross-functional team to orchestrate a campus-wide ideathon for 50+ participants, managing timelines and stakeholder communications.
- **Hackathon Competitor:** Engineered and delivered a fully functional Minimum Viable Product (MVP) within a strict 24-hour time constraint.